

Lab 10: Magnetic Field of a Solenoid

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Pre-lab Questions

1. Use Ampere's Law to develop an expression for the magnetic field $\vec{B}$ inside a solenoid as a function of the number of coils N , the length of the solenoid L , and the current through the solenoid I .

Consider a rectangular Amperian loop C that extends across the entire solenoid, so its horizontal sides span a length L . Its two horizontal sides are parallel to the axis of the solenoid, and its two vertical sides are perpendicular to the axis of the solenoid. One of the horizontal sides is inside the solenoid, and the other horizontal side is outside the solenoid.

By Ampere's Law:

$$\oint_C \vec{B} \cdot d\vec{s} = \mu_0 I_{\text{enclosed}}.$$

Over the entire solenoid, we have N turns of wire, so the total current enclosed by the Amperian loop is $I_{\text{enclosed}} = NI$:

$$\oint_C \vec{B} \cdot d\vec{s} = \mu_0 NI.$$

Since C is piecewise smooth (it is a rectangle), break C up into its four segments:

$$\oint_C \vec{B} \cdot d\vec{s} = \int_{\text{top}} \vec{B} \cdot d\vec{s} + \int_{\text{bottom}} \vec{B} \cdot d\vec{s} + \int_{\text{left}} \vec{B} \cdot d\vec{s} + \int_{\text{right}} \vec{B} \cdot d\vec{s}.$$

There are two important properties of solenoids:

- The magnetic field outside the solenoid is negligible.¹
- The magnetic field inside the solenoid is approximately uniform.

Assume the top side of C is outside the solenoid, and the bottom side of C is inside the solenoid. The magnetic field outside the solenoid is negligible, so the contribution to the line integral from the top side of C is approximately zero:

$$\int_{\text{top}} \vec{B} \cdot d\vec{s} \approx 0.$$

The magnetic field inside the solenoid is approximately uniform, so the contribution to the line integral from the bottom side of C is:

$$\int_{\text{bottom}} \vec{B} \cdot d\vec{s} \approx B \int_{\text{bottom}} ds = BL.$$

Along the left and right sides of C , the magnetic field is perpendicular to the path of integration, so the contribution to the line integral from the left and right sides of C is zero (since dot products of perpendicular vectors are zero):

¹...for an infinitely long solenoid. Since our solenoid is finite, we only make this assumption if we are "vertically" outside the solenoid.

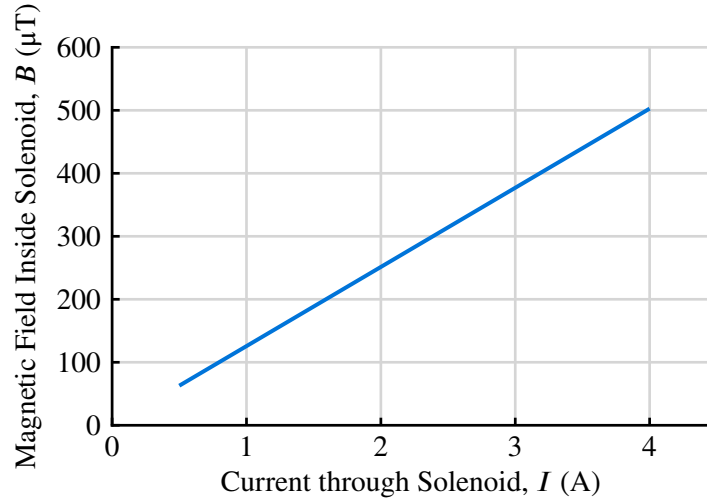
$$\int_{\text{left}} \vec{B} \cdot d\vec{s} = \int_{\text{right}} \vec{B} \cdot d\vec{s} = 0.$$

Thus, we have:

$$\oint_C \vec{B} \cdot d\vec{s} = BL = \mu_0 NI \rightarrow B = \frac{\mu_0 NI}{L}.$$

2. Assuming $N = 30$ and $L = 30$ cm, make a properly labeled plot of the magnetic field B as a function of the current through the solenoid I for $0.5 \text{ A} < I < 4 \text{ A}$.

Magnetic Field inside 30 cm Solenoid with 30 turns at Varying Currents



3. Does the magnetic field B inside the solenoid depend on the position within the solenoid coils?

The magnetic field B inside the solenoid does not depend on the position within the solenoid coils, since the magnetic field inside the solenoid is approximately uniform. This is one of the important properties of solenoids.

4. If the left end of the solenoid lies at $x = 0$, then derive an expression for the magnetic field B along the axis of the solenoid as a function of x , where $x > L$ (outside the solenoid).

If the solenoid has radius R , then it is a distance $r = \sqrt{a^2 + R^2}$ from the current source at $x = L - a$. The Biot-Savart Law gives the magnetic field at x from the current source at $x = L - a$ as:

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{s} \times \hat{r}}{r^2}.$$

Consider just a loop of current at $x = L - a$. $d\vec{s} \times \hat{r}$ is just $\hat{n} ds$ since $d\vec{s}$ is always perpendicular to $\hat{r}$:

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I ds}{r^2} \hat{n}.$$

Splitting $d\vec{B}$ into components:

$$\begin{aligned}
dB_x &= \frac{\mu_0}{4\pi} \frac{I ds}{r^2} \cos \theta \\
&= \frac{\mu_0}{4\pi} \frac{I ds}{a^2 + R^2} \frac{R}{\sqrt{a^2 + R^2}} \\
B_x &= \int_{\text{loop}} dB_x = \int_{\text{loop}} \frac{\mu_0 I}{4\pi} \frac{R}{(a^2 + R^2)^{3/2}} ds \\
&= \frac{\mu_0 I R}{4\pi (a^2 + R^2)^{3/2}} \int_0^{2\pi R} ds \\
&= \frac{\mu_0 I R^2}{2(a^2 + R^2)^{3/2}}.
\end{aligned}$$

$B_y = 0$ due to symmetry.

Since there is a turn density of $n = \frac{N}{L}$ loops, as a ranges from $x - L$ to x , we have:

$$\begin{aligned}
B &= \int_{x-L}^x \frac{\mu_0 I R^2 n}{2(a^2 + R^2)^{3/2}} da \\
&= \frac{\mu_0 I R^2 n}{2} \int_{x-L}^x \frac{1}{(a^2 + R^2)^{3/2}} da \\
&= \frac{\mu_0 I R^2 n}{2} \left(\frac{a}{R^2 \sqrt{a^2 + R^2}} \right) \Big|_{a=x-L}^{a=x} \\
&= \frac{\mu_0 I R^2 n}{2} \left(\frac{x}{R^2 \sqrt{x^2 + R^2}} - \frac{x-L}{R^2 \sqrt{(x-L)^2 + R^2}} \right) \\
&= \frac{\mu_0 I n}{2} \left(\frac{x}{\sqrt{x^2 + R^2}} - \frac{x-L}{\sqrt{(x-L)^2 + R^2}} \right).
\end{aligned}$$

Background Magnetic Field

The magnetic field caused by the Earth and other sources was measured as B_{bg} .

Background Magnetic Field	$B_{bg} = 15 \mu\text{T}$
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Part 1: Magnetic Field as a Function of Current

Number of Turns	$N = 28$
Length of Solenoid	$L = 16 \text{ cm}$
Turn Density	$n = N/L = 1.75 \text{ turns/cm}$

Over 5 trials, different currents I were sent through the solenoid via a DC power supply. For each trial, the magnetic field B_{measured} inside the solenoid was measured using the magnetic field probe.

The correct magnetic field $B_{\text{corrected}}$ was calculated by subtracting the background magnetic field B_{bg} from the measured magnetic field B_{measured} :

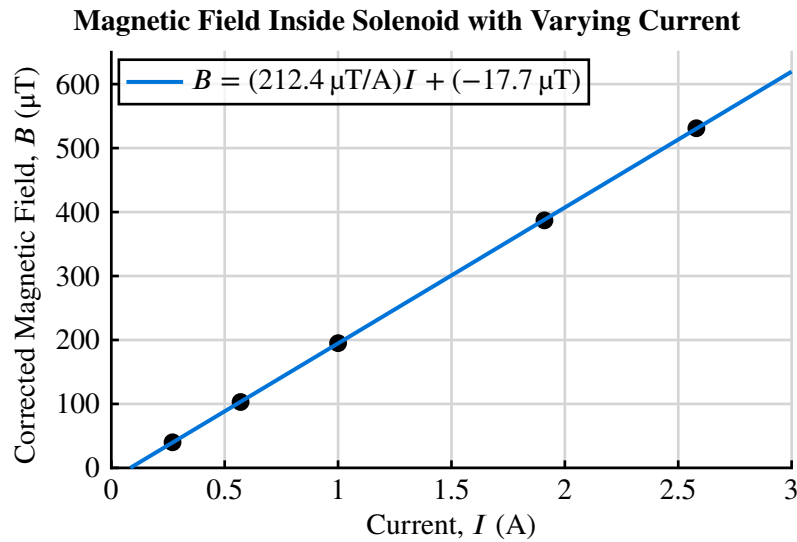
$$B_{\text{corrected}} = B_{\text{measured}} - B_{bg}.$$

The expected magnetic field $B_{\text{calculated}}$ was calculated using the formula derived in the pre-lab:

$$B_{\text{calculated}} = \mu_0 n I.$$

#	Current I	B_{measured}	$B_{\text{corrected}}$	$B_{\text{calculated}}$	% Difference
1	0.27 A	55 μT	40 μT	59 μT	48.44%
2	0.57 A	118 μT	103 μT	125 μT	21.70%
3	1.00 A	210 μT	195 μT	220 μT	12.78%
4	1.91 A	402 μT	387 μT	420 μT	8.54%
5	2.58 A	546 μT	531 μT	567 μT	6.85%

Data Analysis



Performing a linear regression, the best-fit line for $B(I)$ is:

$$B = mI + b, \quad m = 212.4 \mu\text{T/A}, \quad b = -17.7 \mu\text{T}.$$

From the equation $B = \mu_0 n I$, we can see that $m \approx \mu_0 n \rightarrow \mu_0 \approx m/n$:

$$m \approx \mu_0 n \Rightarrow \mu_0 \approx \frac{m}{n} = \frac{2.124 \cdot 10^{-4} \text{ T/A}}{175 \text{ turns/m}} = 1.214 \cdot 10^{-6} \text{ T/(A m)}.$$

The accepted value is $\mu_0 = 4\pi \cdot 10^{-7} \text{ T/(A m)} = 1.257 \cdot 10^{-6} \text{ T/(A m)}$. The percent uncertainty is:

$$\% \text{ uncertainty} = \frac{|\hat{\mu}_0 - \mu_0|}{\mu_0} \cdot 100\% = 3.40\%.$$

To reduce this uncertainty, we could use a more ideal solenoid (i.e., longer) such that the magnetic field inside is more uniform. We could also take a more accurate μ_0 reading of the background magnetic field, since that would cause systematic error in our calculated μ_0 .

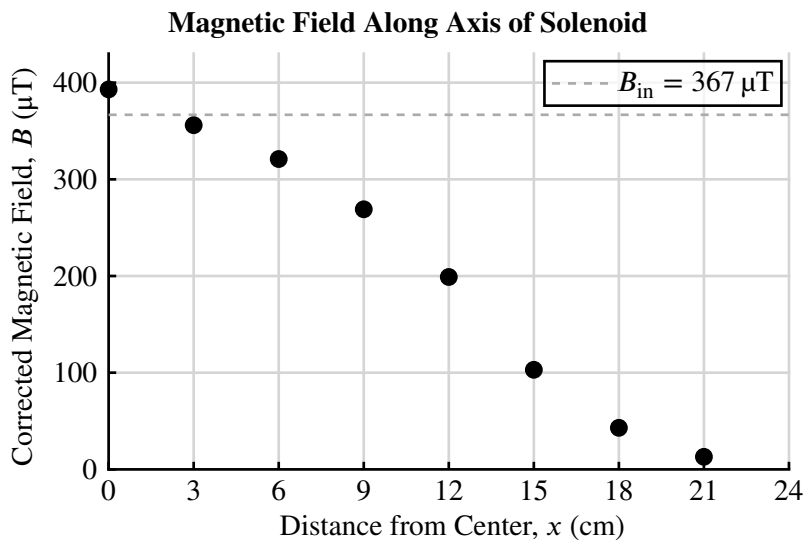
Part 2: Magnetic Field as a Function of Position

Current	$I = 2.71 \text{ A}$
Number of Turns	$N = 28$
Length of Solenoid	$L = 26 \text{ cm}$
Turn Density	$n = N/L = 1.08 \text{ turns/cm}$

The axial distance from the center of the solenoid x was varied across 8 trials. For each trial, the magnetic field B_{measured} along the axis of the solenoid was measured using the magnetic field probe. The corrected magnetic field $B_{\text{corrected}}$ was calculated via $B_{\text{corrected}} = B_{\text{measured}} - B_{\text{background}}$.

#	Distance from Center x	B_{measured}	$B_{\text{corrected}}$
1	0 cm	408 μT	393 μT
2	3 cm	371 μT	356 μT
3	6 cm	336 μT	321 μT
4	9 cm	284 μT	269 μT
5	12 cm	214 μT	199 μT
6	15 cm	118 μT	103 μT
7	18 cm	58 μT	43 μT
8	21 cm	28 μT	13 μT

Data Analysis



The magnetic field **inside** the solenoid (along the central axis) is approximately uniform:

$$B_{\text{in}} = \mu_0 n I.$$

The magnetic field **outside** the solenoid but still along the central axis is (derived in pre-lab Q4, with x_L measured from the left end of the solenoid, L the solenoid length, and R the solenoid radius):

$$B_{\text{out}} = \frac{\mu_0 n I}{2} \left(\frac{x_L}{\sqrt{x_L^2 + R^2}} - \frac{x_L - L}{\sqrt{(x_L - L)^2 + R^2}} \right).$$

These equations don't perfectly agree with my data. Within the solenoid, the magnetic field does not seem to stay constant. Outside the solenoid, the equation better predicts the decreasing behavior of the magnetic field strength, but it creates a discontinuity for $B(x)$ at the solenoid edge.

Plugging in the values for μ_0 , n , and I into the equation for B_{in} , we get a theoretical value of $B_{\text{in}} = 367 \mu\text{T}$. Taking the measured B at the center of the solenoid, it is about $393 \mu\text{T}$, which is about 7.16% different from the theoretical value.

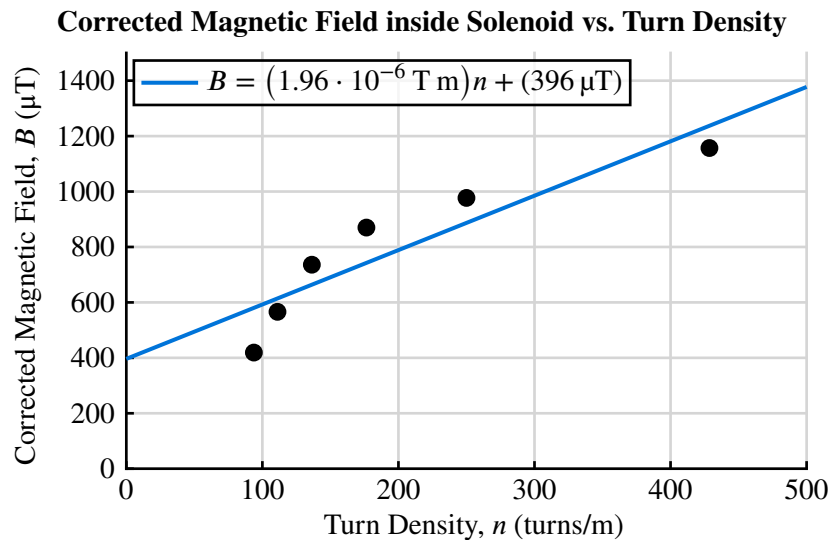
Part 3: Magnetic Field as a Function of Turn Density

Current	$I = 3 \text{ A}$
Number of Turns	$N = 30$

The solenoid remained at 30 turns, while the length of the solenoid L was varied across 6 trials, thus varying the turn density n . For each trial, the magnetic field B_{measured} along the axis of the solenoid was measured using the magnetic field probe. The corrected magnetic field $B_{\text{corrected}}$ was calculated via $B_{\text{corrected}} = B_{\text{measured}} - B_{\text{background}}$.

#	Length L	Turn Density $n = N/L$	B_{measured}	$B_{\text{corrected}}$
1	32 cm	0.94 turns/cm	434 μT	419 μT
2	27 cm	1.11 turns/cm	581 μT	566 μT
3	22 cm	1.36 turns/cm	751 μT	736 μT
4	17 cm	1.76 turns/cm	885 μT	870 μT
5	12 cm	2.50 turns/cm	992 μT	977 μT
6	7 cm	4.29 turns/cm	1172 μT	1157 μT

Data Analysis



Performing a linear regression, the best-fit line for $B(n)$ is:

$$B = mn + b, \quad m = 1.962 \cdot 10^{-6} \text{ T m}, \quad b = 396 \mu\text{T}.$$

Theoretically, the magnetic field inside the solenoid is $B = \mu_0 I n$ with I fixed at 3 A. So, the slope $m = \mu_0 I$. Solving for μ_0 :

$$\mu_0 = \frac{m}{I} = \frac{1.962 \cdot 10^{-6} \text{ T m}}{3 \text{ A}} = 6.541 \cdot 10^{-7} \text{ T/(A m)}.$$

The accepted value is $\mu_0 = 1.257 \cdot 10^{-6} \text{ T/(A m)}$. The percent uncertainty is:

$$\% \text{ uncertainty} = \frac{|\hat{\mu}_0 - \mu_0|}{\mu_0} \times 100\% = 47.9\%.$$

The large percent uncertainty reflects that the data does not follow a perfectly linear $B(n)$ relationship. In the graph, the data points are not close to the line of best-fit, indicating a poor fit (poor r -value). This is likely because our solenoid was far from ideal. We can make our solenoid “more ideal” by increasing the length or decreasing the radius of the solenoid.

Part I is the better method for determining μ_0 because it fixes the probe position at the center of the solenoid, where the magnetic field is most uniform (approximating an ideal solenoid). Part II varies the probe position, including outside the solenoid, which means we have to use a much more complicated equation for B .